



AutoNexa Intelligent Parking and Recall System

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Executive Summary

AutoNexa was established in 2025 by a group of Electrical and Electronics Engineering students at Middle East Technical University with the goal of developing intelligent and sustainable solutions for future mobility. The project focuses on exploring how autonomous driving technologies can enhance everyday transportation tasks through designs that prioritize safety, efficiency, and user convenience. This report presents the conceptual design, system architecture, and experimental validation of the AutoNexa Intelligent Parking and Recall System, a small-scale autonomous vehicle prototype developed to perform autonomous parking and remote recall operations.

The proposed system is designed to operate autonomously in a structured indoor environment, where it identifies a suitable parking space and completes a parking maneuver within a 40 cm × 30 cm area, achieving a positional accuracy of ± 5 cm. In addition to autonomous parking, the vehicle includes a wireless *Summon* feature that allows it to return to a user-defined location upon request. All system functionalities are developed under strict constraints on safety, control latency, and power consumption to reflect practical automotive design considerations. To achieve these objectives, AutoNexa employs a modular system architecture consisting of perception, control, navigation, power management, and wireless communication sub-systems.

Environmental perception is realized through a LiDAR scan and supported by a camera, enabling 3D model of environment, reliable obstacle detection, and parking space identification. Vehicle motion is managed using encoder-equipped DC motors controlled by closed-loop PID algorithms, allowing precise and repeatable maneuvers. High-level decision-making is handled by a finite state machine that coordinates sensor fusion outputs, navigation logic, and user commands to ensure coherent system behavior.

Significant progress has been achieved through systematic sub-system level testing of the localization and navigation components. Sub-system level testing has validated the feasibility of the proposed localization and navigation approach. LiDAR-based experiments demonstrated sufficient position and orientation of accuracy for indoor parking, with average errors remaining within the required limits. Navigation tests confirmed that the system can reliably generate safe paths and reach target locations with high precision under normal operating conditions. However, the experiments also showed reduced performance in confined areas and during long periods of operation, indicating that localization based on a single sensor is not sufficiently reliable. These results motivated the use of sensor fusion and complementary perception methods to improve reliability in the final system design.

The AutoNexa system is currently progressing toward full integration of its sub-systems into a unified autonomous platform. The results presented in this report show that the proposed design is both technically feasible and well justified. Moreover, the planned hybrid sensing and control approach effectively addresses the identified limitations, establishing a solid foundation for complete system integration and final prototype implementation.

Introduction

AutoNexa is a student-driven engineering initiative focused on developing intelligent and sustainable solutions for future mobility applications. The project team consists of Electrical and Electronics Engineering students with experience in control systems, embedded software,

perception, and wireless communication. This report presents the conceptual design of the AutoNexa Intelligent Parking and Recall System, a small-scale autonomous vehicle prototype designed to perform autonomous parking and remote recall operations in structured indoor environments.

The proposed system aims to address limitations commonly encountered in manual parking and conventional driver-assistance solutions, such as imprecise maneuvering, limited situational awareness, and reliance on human judgment in constrained spaces. By integrating camera-based perception, ultrasonic environmental sensing, closed-loop motor control, and wireless communication, the AutoNexa system enables autonomous navigation and precise parking by controlling the vehicle's planar position and orientation. Sensor fusion techniques are employed to combine complementary sensing modalities, improving robustness against individual sensor limitations and environmental uncertainties.

This report is structured to provide a comprehensive overview of the project's conceptual design and current development status. It begins with the motivation behind developing an autonomous parking and recall system, followed by a clear definition of underlying engineering problems. The current state of sub-system development and preliminary testing is then discussed to highlight progress and identify observed limitations. While early experiments focused on individual components, primarily perception and localization-related sub-systems, the final system is designed to integrate multiple sensing and control layers to enable reliable autonomous operation.

The subsequent sections of this report describe the system-level objectives and requirements, the overall system architecture, and the detailed design of the individual sub-systems. Initial test results are also presented to support key design decisions, together with planned integration steps, risk analysis, and validation strategies. By combining systematic system decomposition with experimental evaluation, the report provides a clear roadmap toward the development of a robust and scalable autonomous parking solution.

At least, the AutoNexa project seeks to demonstrate how compact autonomous systems can improve safety, efficiency, and user convenience in everyday transportation scenarios. The conceptual design presented in this report establishes a solid technical basis for full system integration and the realization of the final prototype.

Motivation of the project

The AutoNexa project is motivated by the need for an intelligent, reliable, and user-oriented autonomous parking and recall system that can address the difficulties commonly encountered in constrained parking environments. Parking often requires careful maneuvering, constant attention, and repeated adjustments, which increase driver workload and raise the risk of minor collisions, particularly in indoor or crowded areas. These challenges underline the importance of autonomous solutions that can carry out parking tasks safely and consistently without requiring continuous human involvement.

Many existing parking assistance systems rely on limited sensing capabilities or still require significant manual supervision, which reduces their effectiveness in complex environments. AutoNexa aims to overcome these limitations by integrating multiple sensing modalities and decision-making mechanisms into a compact autonomous platform. The combination of camera-based perception, ultrasonic environmental sensing, and closed-loop control improves spatial awareness and system robustness compared to single-sensor approaches. As a result,

the vehicle is able to autonomously detect obstacles, identify suitable parking spaces, and perform precise parking maneuvers.

In addition to autonomous parking, the system includes a wireless recall (*Summon*) functionality to further enhance user convenience. This feature allows the vehicle to return to a designated location upon request and reflects realistic use cases in which vehicles operate autonomously within short-range, structured environments such as indoor parking areas, laboratories, or test facilities. Implementing and evaluating this capability at a small scale provides practical insight into the challenges associated with autonomous navigation, perception reliability, and system integration.

In addition to its immediate functionality, the AutoNexa project provides an experimental platform for researching autonomous system design under practical engineering constraints, such as imperfect sensing conditions, limited computational resources, and limited power availability. In addition to demonstrating a functional prototype, the goal is to gain a better understanding of how autonomous parking technologies can be improved in terms of accuracy, dependability, and accessibility. The project's experience and knowledge help advance intelligent mobility systems that improve user experience, efficiency, and safety.

Literature/Market Survey (State of the Art)

Literature Survey

Autonomous parking and low-speed navigation have been widely studied in the context of mobile robotics and advanced driver-assistance systems. Prior research emphasizes the importance of sensor fusion to achieve reliable perception and localization in constrained environments. Vision-based methods are commonly used for path detection, landmark recognition, and parking space identification, while range-based sensors such as ultrasonic sensors and LiDAR provide complementary information for obstacle detection and distance estimation.

Research has shown that systems relying on a single sensing modality often experience performance corruption under varying environmental conditions, such as changes in lighting, reflective surfaces, or unsuitable surroundings. Multi-sensor fusion techniques that integrate cameras, ultrasonic sensors, and inertial measurements are frequently used to overcome these difficulties. For structured tasks like autonomous parking, where both the environment and vehicle behavior can be represented using well-defined operational states, architectures based on finite state machines and rule-based navigation have shown promise for decision-making.

In order to guarantee precise and consistent vehicle motion during low-speed maneuvers, closed-loop control strategies—in particular, PID-based motor control—are also frequently used. These methods make it possible to precisely control planar position and orientation, which is essential for autonomous parking in restricted areas.

Market Survey

Autonomous parking and vehicle recall features are becoming more and more common in commercial automotive systems. Automated parking features that decrease driver workload and increase safety in restricted areas are accessible in high-end cars with advanced sensing and computing systems. In order to execute low-speed autonomous maneuvers in structured environments, systems like Bosch's automated parking solutions and Tesla Autopark combine cameras, ultrasonic sensors, and internal processing.

However, these commercial solutions usually go into huge-scale vehicles and depend on exclusive software and hardware, which makes them expensive and difficult to use for testing and instruction. Moreover, a comprehensive investigation of the sensing, control, and decision-making processes is limited by their closed architectures.

The AutoNexa system aims to address these gaps by offering a compact, modular, and cost-effective autonomous parking platform. By focusing on a small-scale implementation and a transparent system design, AutoNexa allows practical experimentation with perception, navigation, and control strategies while still maintaining similarities to real-world automotive systems. This approach makes the project a useful testbed for studying autonomous parking technologies under realistic conditions and helps bridge the gap between academic research and commercial automotive solutions.

Current Status of Project

At the current stage of the project, extensive localization and navigation tests have been successfully conducted using a LiDAR-based sensing framework. The experiments were performed in a controlled indoor environment with a predefined map to evaluate the system's ability to estimate position and orientation accurately and to navigate toward target locations.

Localization Tests:

The LiDAR-based localization subsystem demonstrated reliable performance across most test points. Mean position errors generally remained within a few centimeters, and standard deviation values were low, indicating good repeatability and stability. Orientation estimation also showed consistent results, with mean angular errors typically below the defined acceptance thresholds. However, tests showed that localization accuracy degrades slightly near boundary regions of the map, likely due to environmental geometry and reduced feature asymmetry. Additionally, long-duration tests showed gradual error accumulation over time, indicating LiDAR drift and the need for periodic recalibration.

Navigation Tests:

Navigation experiments showed that the system is able to generate smooth, collision-free paths and reach target positions with acceptable distance errors under normal operating conditions. In successful cases, localization and path planning worked together in a consistent and reliable manner. However, some failure cases were also observed, especially in narrow areas or environments with dense obstacles. In these situations, small localization errors occasionally resulted in invalid or unsafe paths. These observations indicate that navigation performance is highly dependent on localization accuracy and the quality of the map representation.

Overall, the current test results confirm that LiDAR-based localization and navigation are feasible for the project, while also revealing important limitations related to drift, boundary effects, and robustness in challenging environments. Based on these findings, the next phase of the project will focus on improving localization reliability through sensor fusion approaches, such as integrating camera-based marker detection, in order to enhance overall system robustness and navigation accuracy.

Scope and Organization

This report presents the conceptual design and the current development status of the AutoNexa Intelligent Parking and Recall System. It starts with an executive summary that

outlines the main project objectives, key design decisions, and significant findings. The introduction then explains the motivation behind the project and provides an overview of the work completed so far.

Next, the scope and organization of the project are described to give the reader a clear understanding of the report structure. The problem statement section discusses the main engineering challenges related to autonomous parking in constrained environments, while the objectives and requirements section defines the system-level goals, performance targets, and design constraints.

The system design section introduces the overall architecture of the proposed solution and explains how the sub-systems interact with one another, along with the reasoning behind the main design decisions. This section is followed by dedicated sub-system descriptions that provide detailed explanations of individual components, including perception, localization, navigation logic, and supporting infrastructure. Experimental results obtained from sub-system level tests are then presented and analyzed to determine system feasibility and to support the selected design choices.

Finally, the planning section describes the remaining development steps, including integration plans, scheduling, and risk analysis. The report concludes with a summary of the main outcomes and a discussion of the future work required to achieve full system integration and validation.

Problem Statement

The development of the AutoNexa autonomous parking robot addresses the complex challenge of achieving high-precision navigation and maneuvering within a constrained, GPS-denied indoor environment. While autonomous mobility is a well-established field, the specific requirement to execute parking maneuvers with a tolerance of ± 5 cm introduces a set of distinct engineering problems related to state estimation, sensor limitations, control theory, and real-time computational resource management.

The primary engineering problem is achieving precise State Estimation within an enclosed environment lacking absolute positioning references. The robot must primarily rely on "Dead Reckoning"—the integration of relative motion data from wheel encoders and the Inertial Measurement Unit (IMU)—to estimate its trajectory. However, this method is mathematically prone to cumulative drift, where small measurement errors grow unbounded over time. Factors such as high-frequency sensor noise, mechanical backlash in the motor gearboxes, and, most critically, unpredictable wheel slippage on smooth testbed surfaces, cause the estimated position to progressively diverge from the actual physical location. Without a correction mechanism, a robot attempting to maneuver based solely on this internal feedback would inevitably lose track of its position and collide with boundaries or obstacles. Therefore, developing a robust localization architecture that can continuously bound this error by fusing internal states with external environmental data (via SLAM or Visual markers) represents a significant technical hurdle.

A secondary challenge lies in Perception and Sensor limitations. The chosen primary sensor, the RPLIDAR C1, relies on measuring the time-of-flight of laser pulses. While accurate, this technology faces physical limitations, including a minimum detection range (blind spot) and susceptibility to surface reflectivity. In a confined 2x2 m testbed, the robot must maneuver

dangerously close to obstacles, potentially entering the sensor's blind spot (<0.05 m) where it effectively becomes blind. Furthermore, relying on geometric perception alone creates a "Data Association" problem; determining whether a detected gap is a valid parking spot or merely a gap between random obstacles requires robust feature extraction algorithms capable of distinguishing geometric signatures in the presence of sensor noise.

From a Control Engineering perspective, the robot is a Non-Holonomic System. A differential drive chassis cannot move instantaneously in the lateral direction (sideways); it must rotate to change its heading. This imposes strict kinematic constraints on trajectory planning. Generating a smooth, collision-free path that aligns the robot's heading with a parking spot requires continuous curvature planning (e.g., Bézier or Dubins curves). Implementing these control algorithms is further complicated by the non-linear relationship between the commanded motor voltage and the resulting physical velocity, which varies with battery voltage levels and load torque. Designing a closed-loop PID controller that can maintain precise low-speed control without stalling the motors or causing oscillation is critical for the final docking phase.

Finally, the project faces the challenge of Real-Time Embedded Computing. The system must perform simultaneous tasks—processing high-bandwidth LiDAR point clouds, solving SLAM graph optimizations, capturing and processing video frames, and computing safety-critical control loops—all on a single Raspberry Pi 5 Single Board Computer (SBC). Balancing the trade-off between algorithm complexity (accuracy) and processing latency (responsiveness) is vital. A delay in processing sensor data results in the robot reacting to a "past" state of the world, which, at movement speeds of 0.5 m/s significantly increases the risk of collision. Optimizing the software architecture to utilize multi-threading and efficient memory management within the constraints of an embedded OS is a prerequisite for system safety and reliability.

Objectives and Requirements

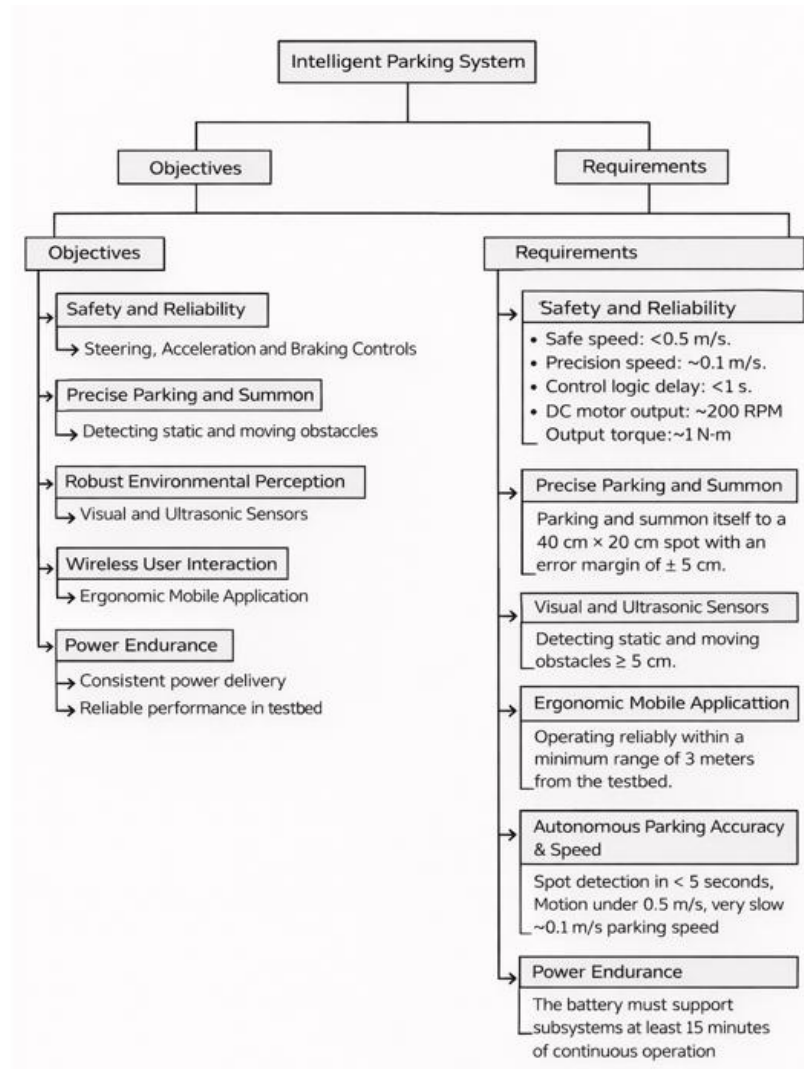


Figure 1. Objectives and Requirements

Objectives

- safe and reliable autonomous operation
- precise parking and summon functionality
- robust environmental perception
- wireless user interaction
- system reliability and endurance

The primary goal of the system is to develop an Intelligent Parking System that can autonomously perform precise parking and summon operations in a controlled indoor

environment. The system is designed to operate safely and reliably, ensuring stable vehicle motion through accurate steering, acceleration, and braking control. Any instability or unsafe behavior during autonomous operation could compromise system performance, making safety and reliability fundamental objectives of the design.

Another key objective is achieving precise parking and summon functionality. The vehicle must be capable of navigating to a designated parking spot and positioning itself accurately within a confined area. In addition, the system should be able to respond to a summon command and return to a predefined location without human intervention. These capabilities demonstrate practical autonomy and highlight the system's ability to perform repeatable, controlled maneuvers.

The system also emphasizes robust environmental perception. Reliable detection of obstacles, boundaries, and parking spaces is essential for autonomous decision-making. By integrating sensing and real-time processing, the system aims to maintain continuous awareness of its surroundings and respond effectively to both static and moving obstacles.

Furthermore, wireless user interaction is a critical objective. Users must be able to issue high-level commands and monitor the system remotely through a wireless interface. The interaction should be intuitive and responsive, ensuring smooth communication between the user and the autonomous vehicle.

Finally, the system is designed with reliability and endurance in mind. All subsystems must operate consistently over extended test sessions, maintaining stable performance without requiring manual intervention or frequent resets.

Requirements

- steering, acceleration, and braking controls
- precise parking and summon operation
- visual and ultrasonic sensing
- wireless mobile application
- battery endurance

The system requires an onboard control unit capable of managing steering, acceleration, and braking in real time. Control logic must ensure smooth vehicle motion at low speeds, enabling precise maneuvering during parking operations while maintaining safe behavior during navigation. The vehicle must operate below a safe speed threshold of 0.5 m/s and support very low-speed motion, around 0.1 m/s, to increase positioning accuracy during parking.

To fulfill the precise parking and summon requirement, the vehicle must autonomously park itself within a 40 cm × 20 cm designated area with an error margin of ± 5 cm. The system must also be capable of returning to a user-defined position upon receiving a summon command, without human assistance. These operations must be completed within a reasonable decision-making time to ensure responsiveness.

The system relies on visual and ultrasonic sensing to achieve robust environmental perception. Sensors must be capable of detecting static and moving obstacles with minimum dimensions of 5 cm in both diameter and height. In addition, the system must be able to detect parking spaces and generate a navigation path within 5 seconds, achieving an overall path and obstacle detection accuracy of approximately 90%.

A wireless mobile application is required to enable user interaction. The vehicle must communicate reliably with the user interface within a minimum range of 3 meters from the testbed. This communication must support command transmission and system monitoring with minimal latency to ensure timely responses.

Finally, the system must satisfy battery endurance requirements. The onboard power system must support all subsystems for at least 15 minutes of continuous operation during a test session. This ensures that the vehicle can complete autonomous tasks and experimental evaluations without unexpected power interruptions.

System Design

Plan A: LiDAR Only

The AutoNexa Autonomous Parking System is designed to execute precise navigation and parking maneuvers in a constrained environment by estimating the vehicle's pose in real-time using geometric data. This compact mobile robot integrates a high precision 2D LiDAR scanner directly onto a differential drive chassis, making it a fully self-contained solution without requiring external global positioning systems or visual fiducial markers. The system achieves robust localization and accurate path planning by leveraging advanced Simultaneous Localization and Mapping (SLAM) algorithms, relying solely on laser ranging data to map the environment and identify parking slots.

The proposed system consists of three main components: the Perception Unit, the Computation Core, and the Mobility Platform. The Perception Unit features a Slamtec RPLIDAR C1, which is rigidly mounted to the center of the chassis to capture 360-degree environmental depth data. The Computation Core, a Raspberry Pi 5, performs real-time signal processing, occupancy grid generation, and trajectory planning. Finally, the Mobility Platform comprises DC motors equipped with optical encoders and a dedicated motor driver interface, which executes the calculated velocity commands to physically move the vehicle within the testbed.

The overall system's functionality begins with the LiDAR capturing planar laser scans to measure the distance to obstacles and walls. This raw point cloud data is sent to the Computation Core, where it is filtered to remove noise and processed using the SLAM Toolbox to construct a dynamic map. By analyzing the geometric signatures within the map—specifically orthogonal wall segments and depth discontinuities—the system identifies the target parking spot. The resulting navigation commands are synthesized into linear and angular velocity vectors and transmitted to motor drivers.

To further enhance the reliability of the robot's localization estimates, the system incorporates a dedicated sensor fusion layer for odometry generation using an Extended Kalman Filter (EKF). By integrating high-frequency tick data from the incremental optical encoders with angular velocity measurements from an onboard Inertial Measurement Unit (IMU), the system

synthesizes a robust dead-reckoning estimate. This sensor fusion algorithm mitigates the inherent limitations of each individual sensor type; it compensates for wheel slippage and uneven terrain using the IMU's gyroscope while simultaneously correcting integration drift in the IMU using encoder feedback. The resulting high-fidelity odometry serves as a critical motion prior for the SLAM algorithm, ensuring map consistency and preventing localization failures during rapid turns or acceleration.

To illustrate the overall hardware configuration, a detailed System Block Diagram is provided in the below System Design Figure. The diagram outlines the interconnections between the LiDAR sensor, the power management subsystem, the Raspberry Pi 5, and the actuation circuit. This system design emphasizes computational efficiency and mechanical simplicity by eliminating the need for complex image processing pipelines, allowing the onboard processor to dedicate its resources entirely to path planning and safety monitoring.

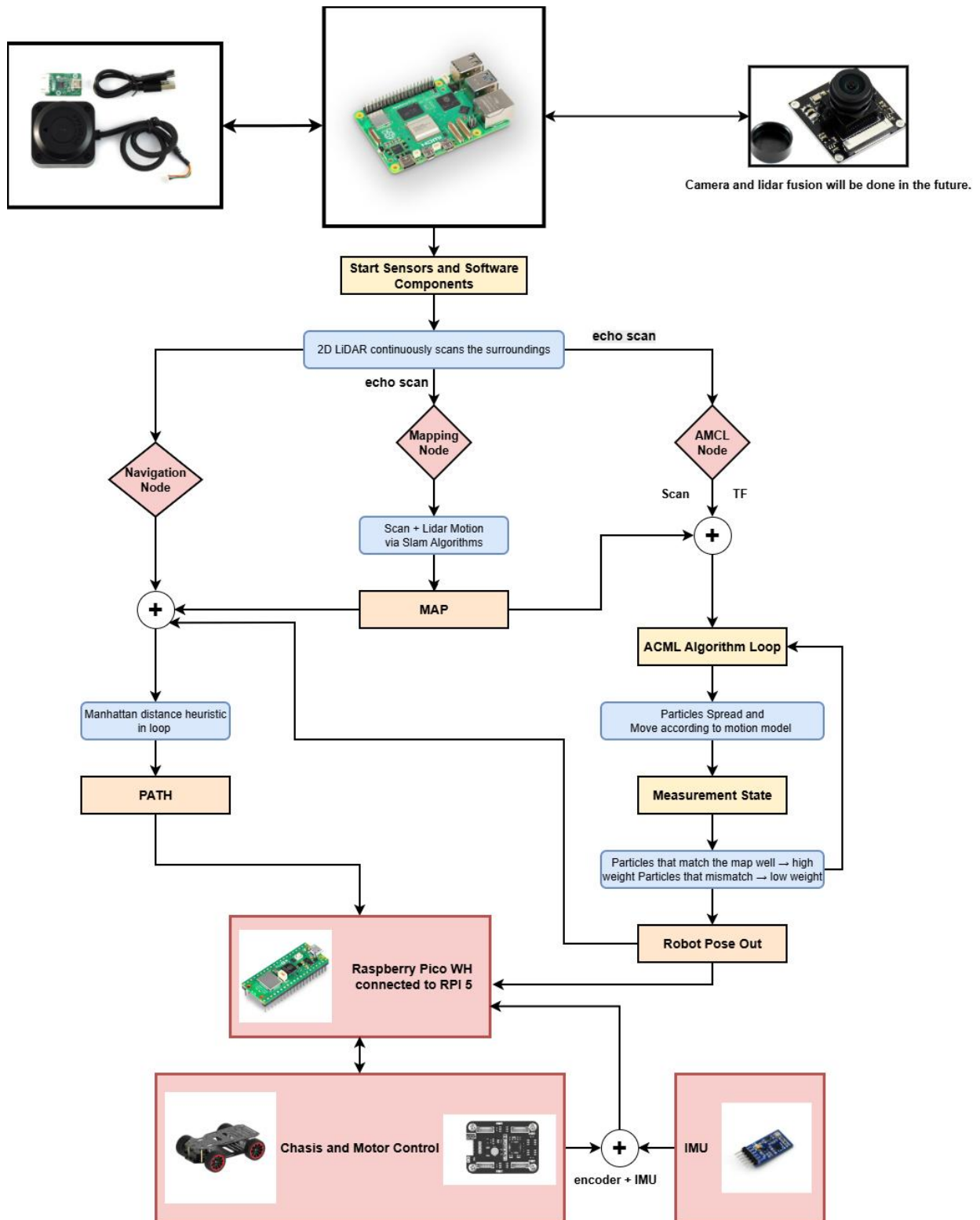


Figure 2. System Design Flow

Additionally, the system is designed to operate in real-time, processing scan data at a frequency of 10-12 Hz, ensuring responsive obstacle avoidance and smooth trajectory

tracking. The design ensures efficient power usage, with the total current draw optimized to allow the 11.1V battery pack to support 1-2 hours of continuous testing. A key feature of this LiDAR-only architecture is its independence from ambient lighting conditions; the active infrared laser source allows the robot to navigate effectively in total darkness or unevenly lit environments, a significant advantage over vision-based systems.

The design decisions for this system were guided by a weighted objective tree, prioritizing environmental robustness, processing latency, and architectural simplicity. Robustness was given high priority, as the system must function reliably regardless of visual texture or lighting changes. Low latency was critical to maintaining safe braking distances, and architectural simplicity ensured that the system remains easy to debug and maintain. This approach provided a systematic method to validate the feasibility of a geometry-first navigation strategy.

The Overall System Design complies with standard robotic safety protocols, including fail-safe collision avoidance zones and emergency stop functionality. It is robust enough to operate in typical indoor environments, leveraging the DTOF (Direct Time-of-Flight) technology of the sensor to minimize blind spots to under 5 cm. Below Figure showcases the physical placement of the LiDAR on the chassis, highlighting the unblocked field-of-view essential for 360-degree mapping.

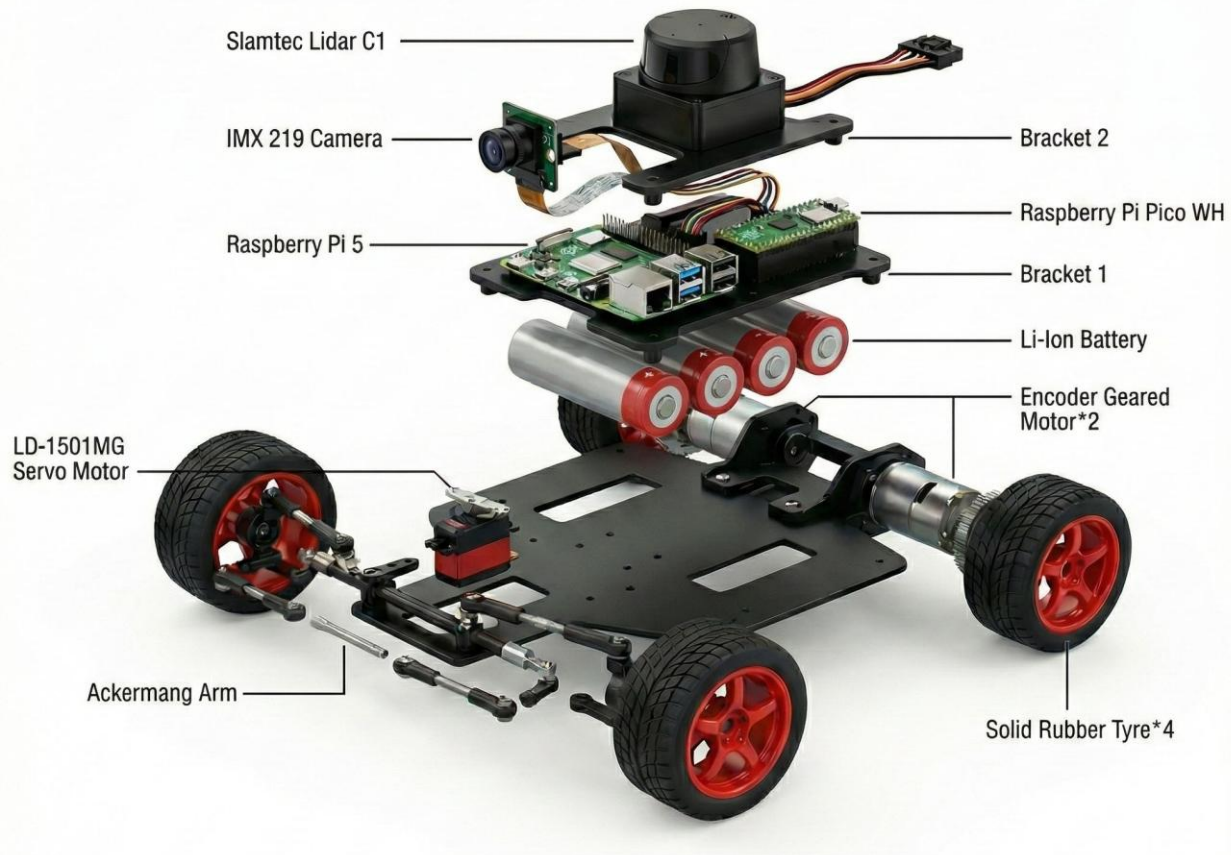


Figure 3. The Model of Product will be built

Sub-system-1 Mapping and Localization

The Mapping and Localization subsystem acts as the primary sensory cortex of the AutoNexa robot. Its function is twofold: first, to construct a consistent 2D Occupancy Grid Map of the unknown environment (Mapping), and second, to estimate the robot's instantaneous pose (x , y , θ) within that map (Localization). This subsystem processes raw laser range data to distinguish between free space, obstacles (walls), and the vehicle's own motion, providing the "State Estimation" required by the Navigation subsystem.

1.1. Requirements Traceability Matrix

The following table outlines the specific requirements derived for this subsystem and maps them to the high-level System Requirements (SR).

Table 1. Requirements traceability matrix for the mapping and localization subsystem.

ID	Subsystem Requirement (SSR)	Description	Related System Requirement (SR)
SSR-1.1	Scan Precision	The subsystem shall resolve geometric features with a minimum resolution of 2 cm.	SR-1: The system must park with an accuracy of ± 5 cm.
SSR-1.2	Update Frequency	The localization estimate shall be updated at a minimum rate of 10 Hz to support control loops.	SR-2: The system must operate in real-time at speeds up to 0.5 m/s.
SSR-1.3	Blind Spot Minimization	The sensor must detect obstacles as close as 0.05 m to ensure safe maneuvering in tight parking spots.	SR-3: The system must avoid collisions in a constrained 2x2 m testbed.
SSR-1.4	Map Consistency	The generated map must remain consistent (no double walls) after multiple passes (Loop Closure).	SR-4: The system must operate autonomously for extended durations without position drift.
SSR-1.5	Processing Latency	The Scan-to-Map matching algorithm must consume less than 30% of the CPU resources on the Raspberry Pi 5.	SR-5: The system must be compact and self-contained (Single Board Computer).

1.2. Main Solution: Sensor Fusion & Graph-Based SLAM

The primary solution employs a hierarchical architecture that first stabilizes the robot's motion estimate using Sensor Fusion and then corrects that estimate using Graph-Based SLAM (SLAM Toolbox). This two-stage approach was selected to decouple the high-frequency/short-term accuracy of inertial sensors from the low-frequency/long-term accuracy of the LiDAR.

Key Algorithms & Process Flow:

1. Stage 1: Dead Reckoning (EKF Sensor Fusion):

Before the SLAM algorithm processes any laser data, the subsystem generates a high frequency (50Hz) "Odometry" estimate using an Extended Kalman Filter (EKF) via the robot_localization package.

- Inputs:

- Wheel Encoders: Provide linear velocity (v_x) based on wheel ticks. Susceptible to wheel slip.
 - IMU (Inertial Measurement Unit): Provides angular velocity (w_z) and linear acceleration (a_x, a_y). Susceptible to drift over time.
 - Algorithm: The EKF fuses these noisy inputs by predicting the state based on the kinematic model and correcting it with incoming measurements. It specifically uses the IMU's gyroscope to compensate for the rotational errors common in differential drive robots (e.g., one wheel slipping on a smooth floor), resulting in a robust odom to base_link transform.
2. Stage 2: Scan Matching (SLAM Front-End): The SLAM Toolbox receives the fused odometry from Stage 1 as a "motion prior" (a guess of where the robot moved). It then uses the Slamtec RPLIDAR C1 data to refine this guess.
- Algorithm:* Karto Scan Matcher. It uses the "Correlative Scan Matching" technique to align the current laser scan with the existing map. Because the EKF has already provided a good initial guess, the scan matcher converges much faster and with higher accuracy than if it relied on laser data alone.
3. Stage 3: Pose Graph Optimization (SLAM Back-End): As the robot explores, small residual errors in the EKF and scan matching can accumulate. To solve this, the system builds a "Pose Graph":
- Nodes represent the robot's pose at specific timestamps.
Edges represent spatial constraints (measurements) between these poses.
- Loop Closure:* When the robot revisits a known area, the algorithm detects the geometric similarity in the laser scan and adds a "Loop Closure" constraint. A solver (Sparse Pose Adjustment) then minimizes the error across the entire graph, correcting the map retroactively and eliminating the drift accumulated by the IMU and Encoders.

Sub-system-2-Navigation

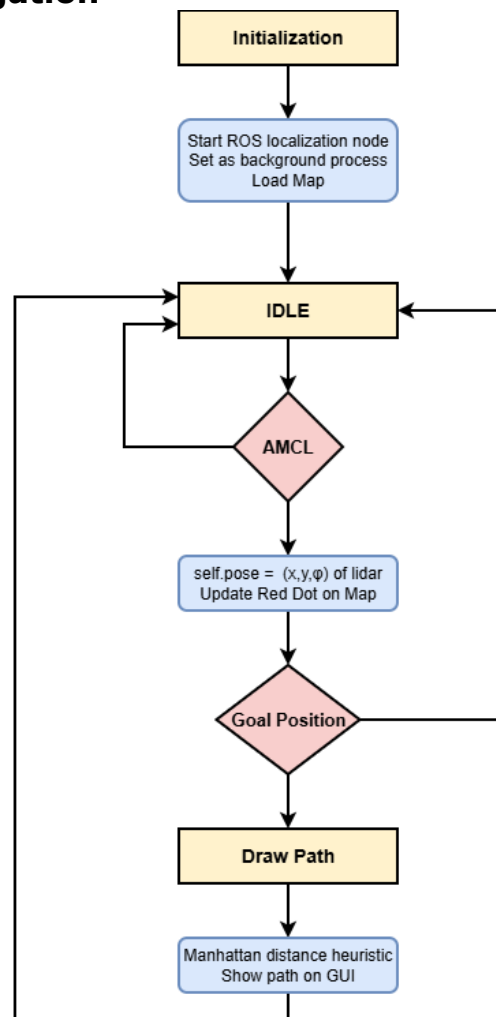


Figure 4. Navigation subsystem architecture and data flow.

The navigation sub-system is a hybrid structure of ROS2 and Python visualization layer. The program listens to the AMCL node (localization) as a background thread. When there is change in movement, the data coming from AMCL node is transformed to previously saved map's coordinates system to visualize correctly. After updating the pose of a robot on a map, it checks whether there is a goal parking spot. If there is goal pose, the path is drawn dynamically by using Manhattan distance heuristic. Note that, since AMCL node is a daemon thread, it does not freeze the python GUI.

Manhattan Distance Heuristic: Manhattan distance assumes you can only move along grid lines (horizontally and vertically), much like a taxi driving through city blocks. It actually calculates the estimated cost from the current node to the goal node by summing the absolute differences of their Cartesian coordinates in map base

$$|x_{current} - x_{goal}| + |y_{current} - y_{goal}|$$

Sub-system-3 Control

The control subsystem of the AutoNexa Intelligent Parking and Recall System is responsible for generating smooth, stable, and precise vehicle motion during low-speed navigation and autonomous parking maneuvers. The vehicle is built on an Ackermann steering chassis, which closely resembles the steering geometry of real passenger vehicles. This makes the control design both realistic and suitable for autonomous parking scenarios.

Ackermann Steering Model

Unlike differential-drive robots, the Ackermann chassis uses a front-wheel steering mechanism combined with rear-wheel propulsion. During a turn, the inner and outer front wheels follow circular paths with different radii. To avoid wheel slippage and ensure smooth turning, the steering angles must satisfy the Ackermann steering condition, where all wheels intersect at a common instantaneous center of rotation.

In the AutoNexa system, the steering angle is controlled via a servo motor, while forward and backward motion is generated using a DC motor connected to the rear axle. By adjusting the steering angle and propulsion speed simultaneously, the vehicle can execute smooth curved trajectories, which are essential for precise parking maneuvers in confined spaces.

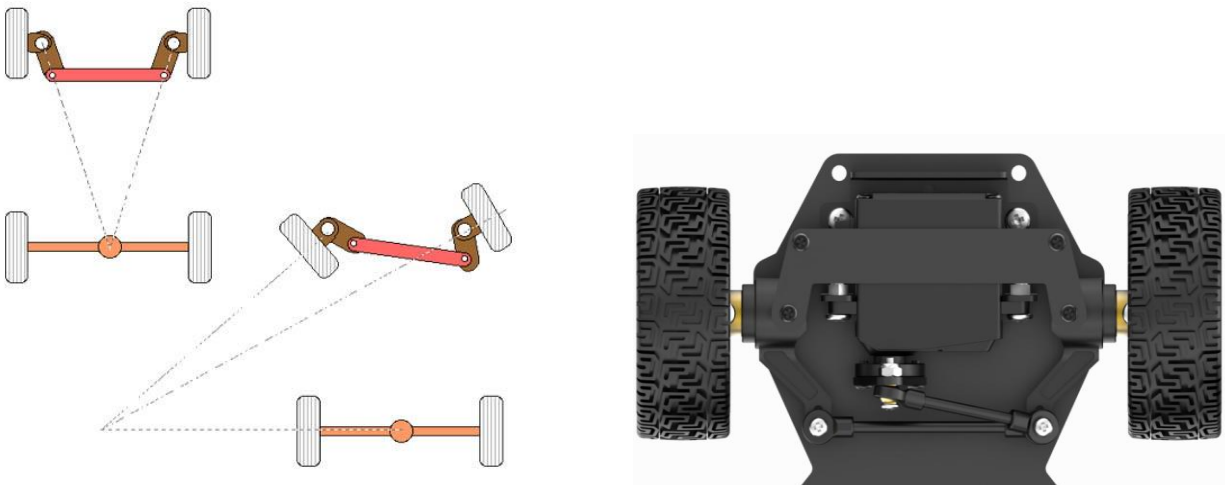


Figure 5. Ackermann Steering

Motion Control Strategy

The motion control strategy is designed for low-speed operation, prioritizing accuracy and stability over high-speed performance. The vehicle's motion is controlled using two primary commands:

- Linear velocity command, which determines the forward or backward speed of the vehicle

- Steering angle command, which defines the turning radius of the vehicle

These commands are generated by the navigation and decision-making module based on the current vehicle state and the desired target position. The control subsystem converts these high-level motion commands into low-level actuator signals for the DC motor and the steering servo.

To ensure predictable and repeatable behavior, acceleration and deceleration are applied gradually, avoiding sudden changes that could cause wheel slip or control instability. This approach is particularly important during parking, where small position errors can significantly affect final accuracy.

Velocity and Steering Control

Vehicle speed is controlled by driving the DC motor through a motor driver using pulse-width modulation (PWM). The system is designed to operate at low speeds so that the vehicle can make small and precise movements when positioning itself in a parking space.

Steering is controlled using a servo motor that adjusts the front wheel steering angle. Mechanical limits are applied to the steering range to avoid excessive turning that could cause instability or violate the Ackermann steering geometry. By coordinating speed and steering commands, the vehicle is able to move smoothly along both straight and curved paths.

Although the control system supports closed-loop feedback using wheel encoders, the current proposal phase focuses on defining the motion model and control structure. Encoder-based feedback and more advanced control methods are planned for later stages to improve system accuracy and robustness.

Relevance to Autonomous Parking

The Ackermann-based motion control design is suitable for autonomous parking because it follows the same movement principles as real vehicles. This allows the system to perform realistic parking actions, such as reverse parking and step-by-step alignment, even in narrow spaces.

By using Ackermann steering and focusing on low-speed motion control, AutoNexa achieves a simple and reliable control structure. This approach also makes it easier to integrate perception and navigation modules, while keeping vehicle motion stable and predictable during autonomous operation.

PID:

In this project, PID control is a critical component for achieving reliable and predictable motion of the vehicle. There are several factors that can affect the speed of DC motors, such as the load, the ground friction, the battery voltage, and the mechanical tolerances. Without closed-loop control, a fixed PWM or voltage command would result in inconsistent wheel speeds, leading to poor trajectory tracking, unstable vehicle motion, and accumulated positioning errors. Encoder feedback and PID control allow the controller to automatically compensate for

disturbances and nonlinearities by continuously measuring the actual motor speed and comparing it to the desired reference speed. By doing this, the wheels maintain their commanded speed under varying operating conditions, including acceleration, deceleration, and slope changes. In this project, motor speed control is especially crucial for higher-level tasks such as localization, navigation, and sensor-based decision making, where consistent motion is required for reliable perception and control.

The simulink block diagram in Figure 6 shows the PI control algorithm of the estimated plant (dc motor) transfer function based on our DC motor specifications (110 rpm at 12V, 15 kg.cm torque, 1:90 gear ratio). The set point is selected as 0.1 m/s (25 rpm) and the step response of the motor with the tuned PI parameters ($K_P = 0.2$, $K_I = 1.32$) is shown in Figure 7. By conducting experiments with the motor and performing bump tests, a better estimated transfer function will be found, and the PI parameters will be recalculated.

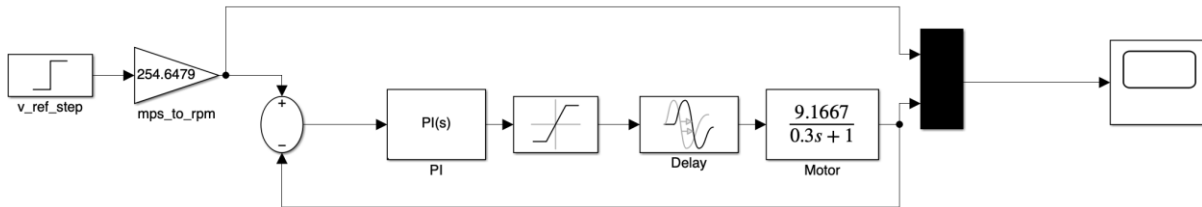


Figure 6. Simulink Block Diagram of DC Motor Control

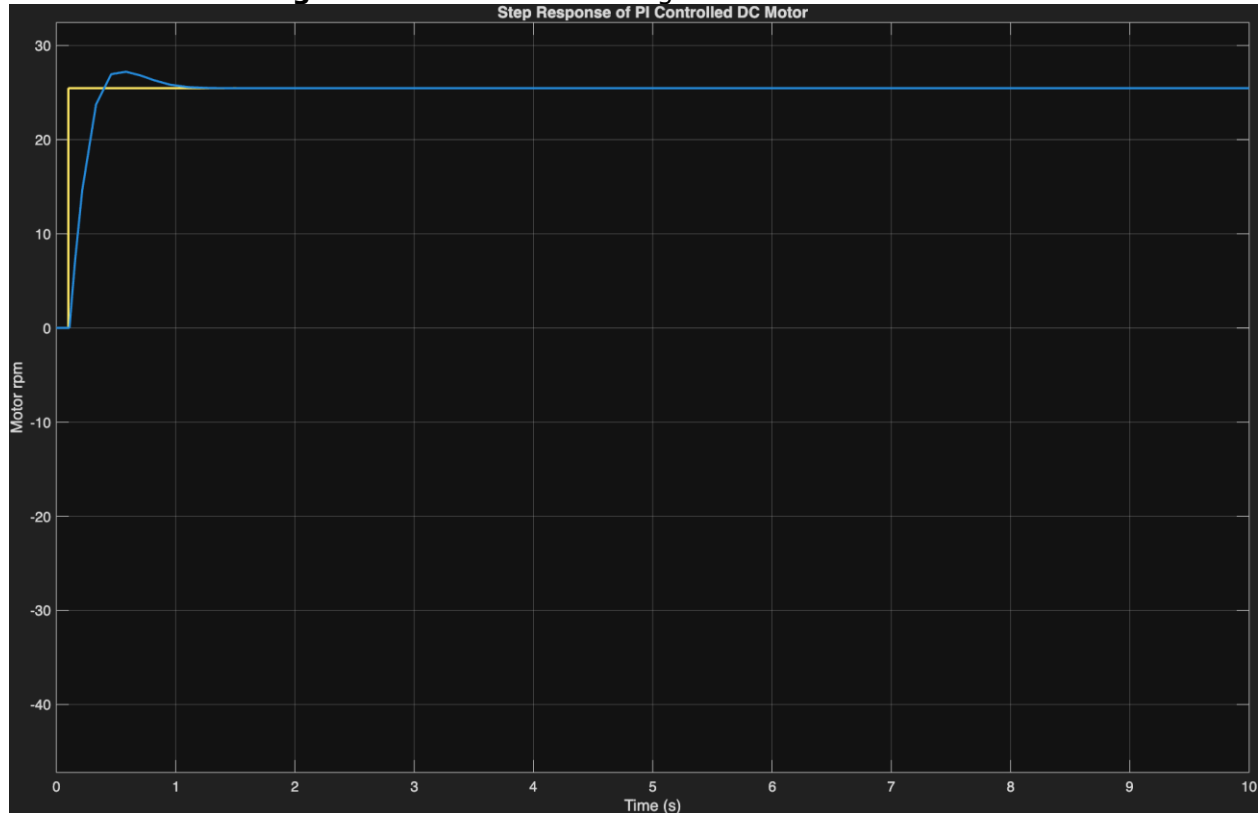


Figure 7. Simulated Step Response of the DC Motor

Sub-system-4 Power System

The power and battery subsystem is one of the most critical subsystems of the autonomous vehicle, as it directly affects the reliability, stability, and overall performance of the entire system. All computation, sensing, and actuation components depend on a properly designed power architecture to operate within their specified voltage and current limits. An insufficient or unstable power supply could lead to sensor malfunction, computation errors, or loss of vehicle control. Therefore, a robust and well-structured power subsystem is essential for safe and reliable autonomous operation.

A centralized power architecture is adopted in this project, where a single battery pack acts as the main energy source and distributes power to all subsystems through regulated voltage levels. The overall structure of the power and battery subsystem and the energy flow between components are illustrated in Figure 8, which presents the power system block diagram.

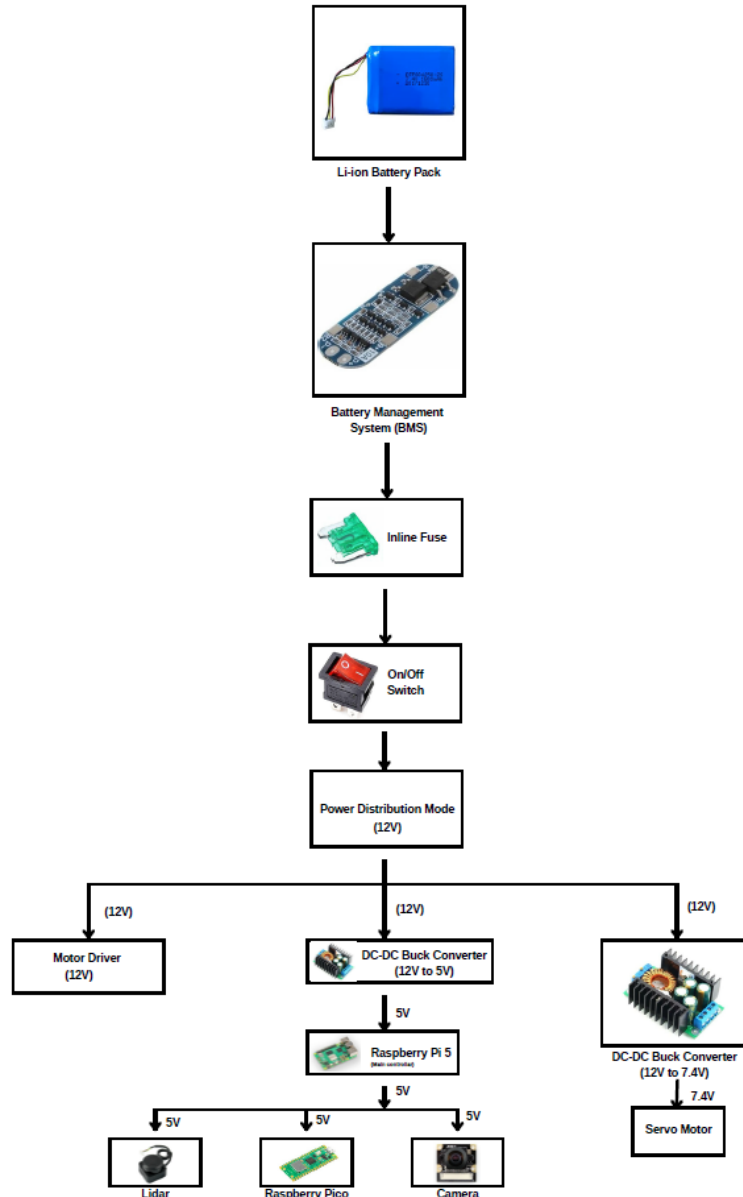


Figure 8. Power system block diagram and energy distribution.

Lithium-ion batteries based on 18650 cells were selected due to their high energy density, rechargeability, and widespread use in mobile robotic platforms. The battery pack consists of three 18650 cells connected in series, forming a 3S configuration with a nominal voltage of 11.1 V and a fully charged voltage of 12.6 V. This voltage level allows the propulsion subsystem to be powered directly while also enabling efficient voltage regulation for lower-voltage electronics.

To ensure safe and reliable operation, the battery pack is integrated with a Battery Management System (BMS). The BMS provides protection against overcharging, over-discharging, and overcurrent conditions, and also performs cell balancing to maintain uniform voltage levels across the battery cells. After the BMS, an inline fuse and a manual ON/OFF switch are placed on the main power line. This configuration protects the system against

electrical faults and allows the entire vehicle to be shut down quickly during testing and debugging.

As shown in Figure 8, power is distributed to different subsystems according to their voltage requirements. The motor driver is supplied directly from the battery voltage, as the DC motors are designed to operate within the nominal 12 V range. For the low-voltage electronics, a DC–DC buck converter is used to step down the battery voltage to a regulated 5 V supply. This regulated output powers the Raspberry Pi 5, which serves as the main computation and control unit of the system. The sensing devices, including the 2D LiDAR and the camera module, as well as the auxiliary Raspberry Pi Pico, are powered through the Raspberry Pi via USB connections.

The Raspberry Pi 5 is assumed to operate under high computational load, which is expected in an autonomous system running perception and localization algorithms. In this operating mode, the Raspberry Pi 5 may require up to 5 A at 5 V, corresponding to a power consumption of:

$$P_{RPI} = 5 \text{ V} \times 5 \text{ A} = 25 \text{ W}$$

$$P_{sensors} = 7.5 \text{ W}$$

The sensing subsystem consists of a 2D LiDAR and a camera module. Based on typical datasheet values, the LiDAR is estimated to consume approximately 5 W, while the camera consumes approximately 2.5 W. Therefore, the total power consumption of the sensing subsystem is estimated as:

The steering subsystem is supplied by a separate DC–DC buck converter that regulates the battery voltage down to approximately 7.4 V. Under nominal operating conditions, the steering servo motor is expected to draw approximately 1.5 A, resulting in an estimated power consumption of:

$$P_{servo} = 7.4 \text{ V} \times 1.5 \text{ A} \approx 11 \text{ W}$$

The propulsion subsystem consists of DC motors powered directly from the battery. During low-speed operation, which aligns with the project requirements, the average power consumption of the motors is estimated to be approximately 10 W. Short-duration peak currents during acceleration are not included in this average estimation, as they do not significantly affect the overall energy budget.

Based on these estimations, the total average power consumption of the system is calculated as:

$$P_{total} = 25 + 7.5 + 11 + 10 \approx 53.5 \text{ W}$$

The total energy capacity of the battery pack is calculated using the nominal voltage and capacity:

$$E_{battery} = 11.1 \text{ V} \times 2.5 \text{ Ah} \approx 27.8 \text{ Wh}$$

Using this energy capacity, the expected continuous operation time of the system is estimated as:

$$t = \frac{27.8}{53.5} \approx 0.52 \text{ hours}$$

This corresponds to approximately 35–40 minutes of continuous operation under high computational load. This duration satisfies the minimum runtime requirement of the project for experimental testing and system validation.

In summary, the proposed power and battery subsystem ensures that all components operate within their required voltage and current limits while providing sufficient runtime for the intended use cases. The block diagram presented in Figure 8 and the accompanying power budget analysis demonstrate that the selected battery configuration and power distribution architecture are adequate and well-suited for the autonomous vehicle system.

Sub-system-5 Mobile App and Wireless Control

Overview

The Mobile Application and Wireless Control Subsystem serves as the primary Human-Machine Interface (HMI) for the AutoNexa platform. Its primary objective is to untether the operator from the physical robot, allowing for real-time remote monitoring, telemetry analysis, and high-level command input over a local wireless network. This subsystem ensures that the complex data processed by the onboard computer is visualized in a user-friendly format on a mobile device.

Methodology and Architecture

The system is built upon a Client-Server architecture.

- **Server Side (Robot/PC):** The main navigation algorithm runs a lightweight web server (using Flask) directly on the robot's onboard computer. This server captures the processed video frames—which include the ArUco path planning overlays, SLAM visualization, and telemetry data—and encodes them for low-latency network transmission (MJPEG streaming).
- **Client Side (Mobile App):** A cross-platform mobile application, developed with Flutter, acts as the client. It connects to the robot's IP address via Wi-Fi to render the live data stream and send asynchronous control commands back to the robot.

Key Features

The application provides a centralized dashboard for the following functionalities:

- **Real-Time Path Finding Visualization:** The app mirrors the computer vision output, displaying the live camera feed overlaid with ArUco marker detection and the calculated Bézier curve paths. This allows the operator to verify the path planning algorithm visually in real-time.
- **LiDAR and SLAM Visualization:** Alongside the camera feed, the interface renders LiDAR mapping data, showing the robot's perception of obstacles and boundaries within the environment.
- **Wireless Target Control:** The operator can dynamically change the robot's objective using the Target ID Selection feature. This allows for switching between different ArUco markers (destinations) remotely without interrupting the autonomous loop.

- Remote Calibration: The system includes a wireless calibration interface, enabling the user to input real-world distance reference values (in cm) directly from the phone to ensure the pixel-to-metric conversion remains accurate.
- Telemetry Dashboard: A dedicated status bar monitors critical system health parameters, including:
 - Battery Level: Real-time voltage monitoring to prevent power loss during operation.
 - Velocity: Instantaneous speed of the mobile platform.
 - Heading/Angle: Current orientation relative to the target.

By offloading the visualization to a mobile device, this subsystem significantly increases the testing efficiency and usability of the AutoNexa project, providing a seamless bridge between complex autonomous logic and human supervision.

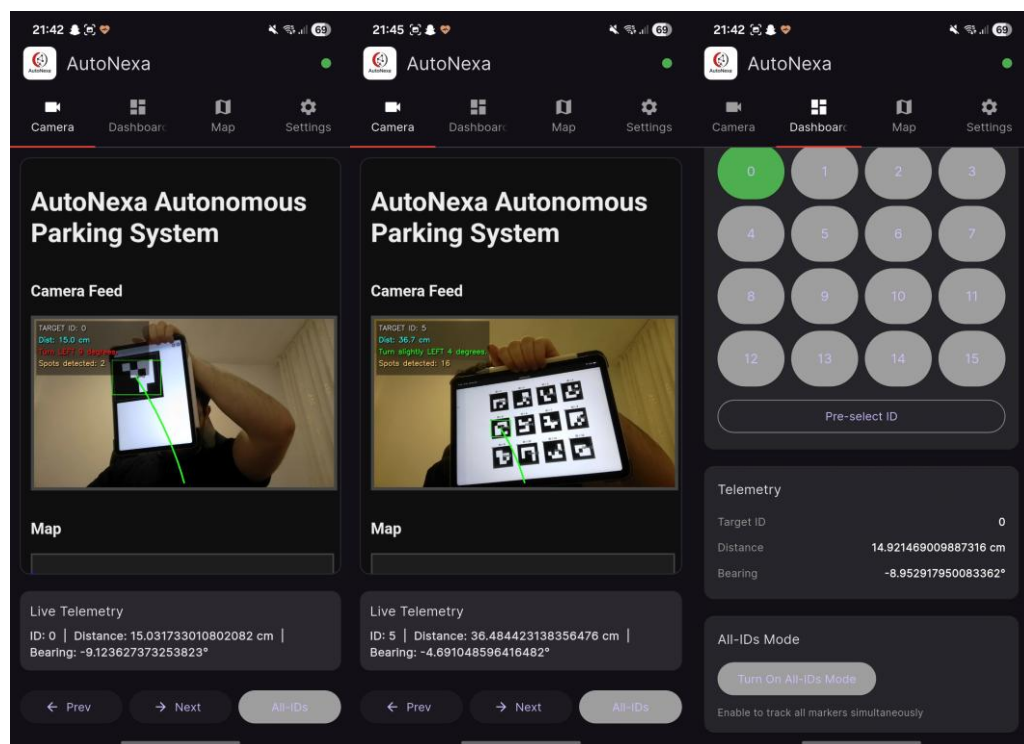


Figure 9. Screenshots Inside the Mobile App

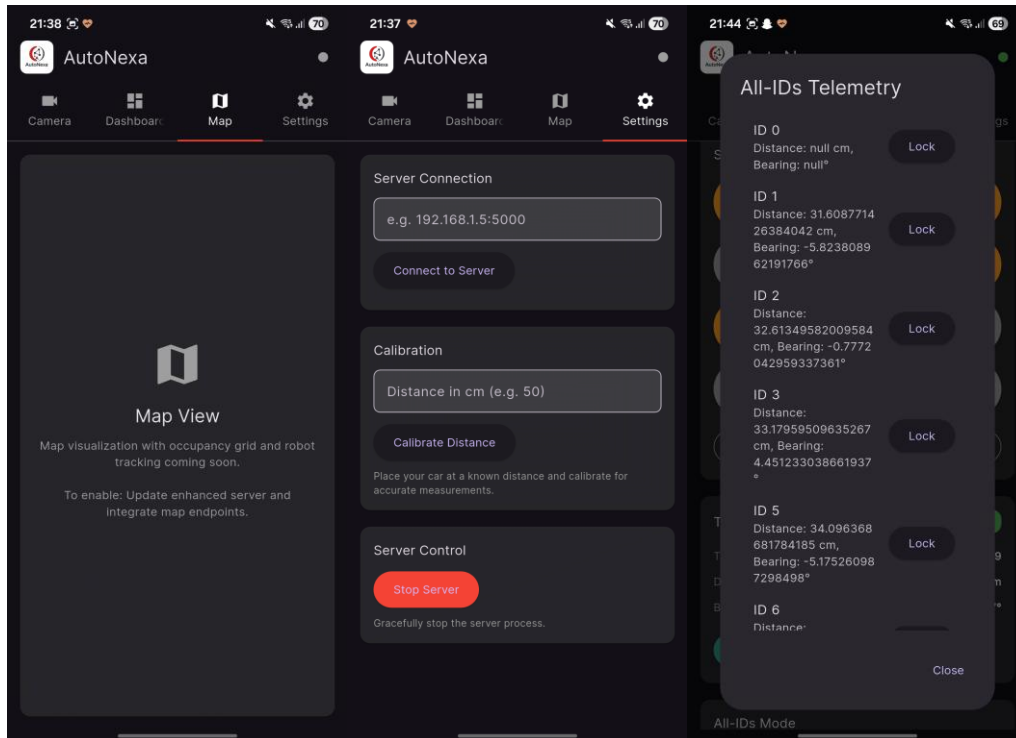


Figure 10. Screenshots Inside the Mobile App:

Plan B: Sensor Fusion (Hybrid LiDAR + Vision)

Description:

The alternative approach implements a Multi-Modal Sensor Fusion architecture, combining the RPLIDAR C1 with a Camera. In this configuration, the sensors are assigned specialized roles based on their strengths: the LiDAR is responsible for Macro-Navigation (Global Localization and Obstacle Avoidance) via SLAM, while the Camera is dedicated to Micro-Alignment (Terminal Docking) via Visual Servoing. The system uses Aruco markers placed at parking spots to provide semantic information (Parking ID) and precise 6DoF pose estimation relative to the target. The navigation logic switches from the global planner to a visual feedback loop when the robot approaches the target vicinity, using the camera to fine-tune the final parking maneuver.

Reasons for Selection:

This plan serves as a robust contingency to address the limitations of geometric recognition. In scenarios where parking spots lack distinct geometric features (e.g., flat walls marked only by paint) or where the LiDAR resolution is insufficient at long ranges, the LiDAR-only approach may fail to identify the target. Sensor fusion is necessary to meet the strict project requirement of ± 5 cm accuracy, as visual markers provide a definitive "anchor" point that allows the robot to correct cumulative odometry errors (drift) that naturally occur during LiDAR-based navigation.

Table 2. Comparison of Plan A (LiDAR-only) and Plan B (LiDAR + Vision).

Feature / Criteria	Plan A: LiDAR-Only Navigation	Plan B: Sensor Fusion (LiDAR + Vision)
Primary Perception Sensor	Slamtec RPLIDAR C1 (2D Laser Scanner)	RPLIDAR C1 + Camera
Navigation Philosophy	Geometric: Relies purely on the physical shape of the environment and obstacles.	Multi-Modal: Combines geometric mapping with semantic visual anchors.
Parking Spot Detection	Feature Extraction: Detects "U-shaped" orthogonal wall segments and depth discontinuities.	Fiducial Tracking: Detects AruCo markers (IDs) placed at the parking spot.
Terminal Docking Method	Open-Loop / Map-Based: Navigates to a coordinate on the map. Accuracy depends on localization quality.	Closed-Loop Visual Servoing: Continuously corrects heading and lateral error based on real-time camera feedback.
Localization Precision	Susceptible to cumulative drift over time. Accuracy is typically ± 10 cm.	Corrects drift at the target location. Accuracy can reach ± 5 cm or better.
Lighting Sensitivity	Robust: Active infrared laser works in total darkness, shadows, or variable lighting.	Sensitive: Requires adequate ambient lighting; performance degrades in low light or glare.
Semantic Capability	None: Cannot distinguish between identical parking spots (e.g., Spot 1 vs. Spot 2).	High: Can read unique IDs to differentiate specific parking assignments.
Computational Load	Low: Processing 2D point clouds is computationally inexpensive for the Raspberry Pi 5.	High: Requires simultaneous processing of Point Clouds and High-Res Video Frames.
System Complexity	Moderate: Single sensor pipeline; no extrinsic calibration required.	High: Requires time synchronization and extrinsic calibration between Camera and LiDAR.

Feature / Criteria	Plan A: LiDAR-Only Navigation	Plan B: Sensor Fusion (LiDAR + Vision)
Failure Modes	May fail if the parking spot lacks distinct walls (e.g., lines painted on floor).	May fail if the marker is occluded, damaged, or if the environment is too dark.

Test Results

This section presents the key findings from the testing of the localization and navigation subsystems of the Intelligent Parking System, specifically evaluating the LiDAR-based perception and autonomous path planning capabilities.

Localization Test

The localization system was tested at 12 predefined grid positions with 8 different orientations and 3 repetitions per position. Pass conditions: mean position error ≤ 10 cm, maximum error ≤ 20 cm, standard deviation ≤ 8 cm, and mean orientation error $\leq 15^\circ$.

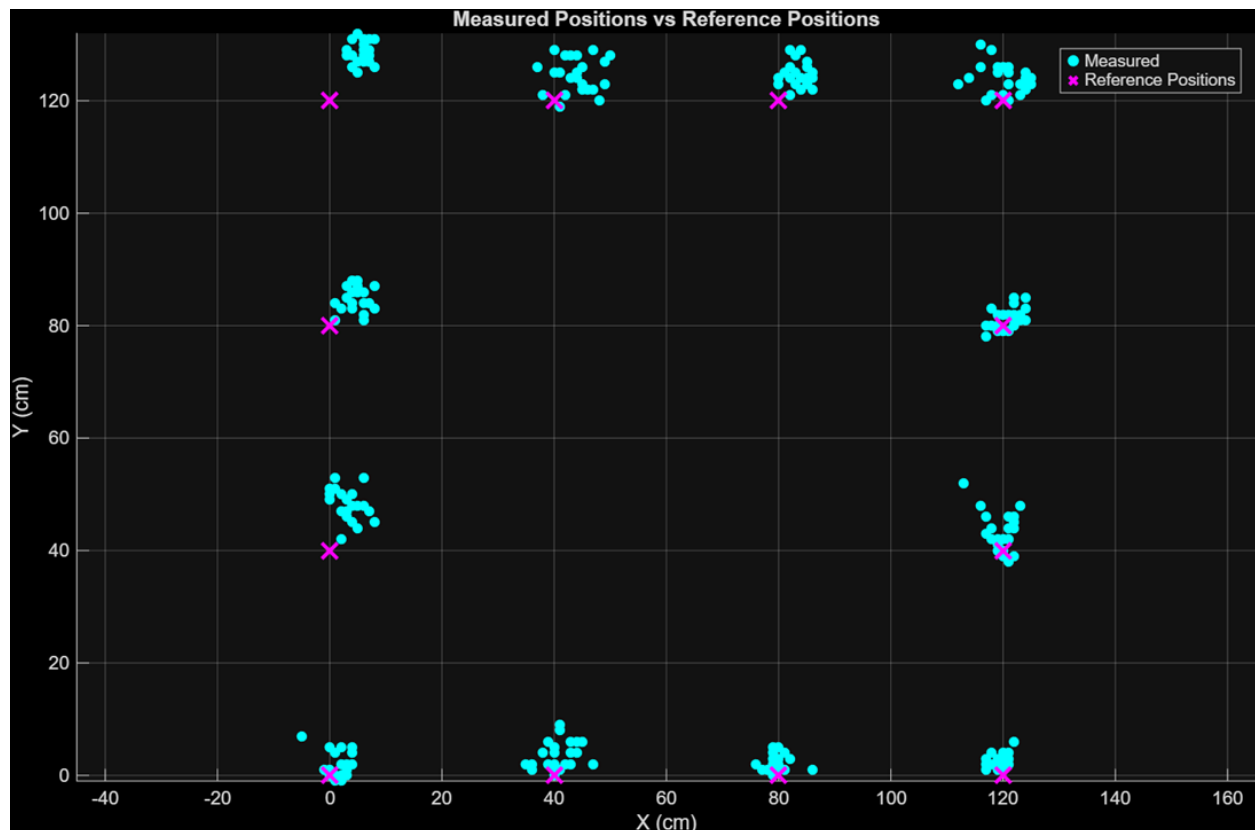


Figure 11. Measured vs. Reference Positions in the Test Area

Position Accuracy

Challenge: Performance varied significantly between central and boundary locations.

Key Findings:

- Mean position errors: 3.00–10.28 cm (all within ≤ 10 cm acceptance threshold)
- Central positions showed lower errors (3.00–3.07 cm); boundary positions higher (9.33–10.28 cm)
- Standard deviation consistently low (1.15–3.16 cm), indicating good repeatability
- Maximum errors ranged 6.08–14.32 cm, mostly below 20 cm limit

Assessment: All positions met acceptance criteria; boundary locations approach the limit due to sensor geometry effects.

Heading Estimation

Key Findings:

- Mean angle errors: 1.99–4.56° (well below 15° threshold)
- Standard deviation: 0.89–2.99°
- Maximum angle errors remained below 15° across all positions

Assessment: Orientation estimation exceeded requirements with consistent performance.

Temporal Drift

Critical Finding:

Table 3. Localization drift over time during long-duration operation.

Time	Error Increase
10 min	1%
30 min	7%
1 hour	10%

Progressive drift indicates periodic recalibration needed for operations beyond 20-30 minutes. Therefore, LiDAR can calibrate itself automatically by using globalization after every parking action which ensures proper operation without drift.

Navigation Test



Figure 12. Navigation was tested across 5 target locations (IDs 8, 7, 13, 14, 15) from multiple starting positions. Success criterion: distance error ≤ 8 cm without obstacle collision.

Performance by Target

Challenge: Systematic failures concentrated at specific spatial regions.

Key Findings:

- **ID 8:** 4.08–7.93 cm errors, zero failures (100% success)
- **ID 7:** 4.84–6.22 cm errors, zero failures (100% success)
- **ID 13:** 4.83–6.16 cm errors, zero failures (100% success)
- **ID 14:** Failures at positions (120,120) and (120,80) - 83.3% success rate
- **ID 15:** Identical failures at (120,120) and (120,80) - 83.3% success rate

All successful navigations met the 8 cm acceptance threshold.

Failure Root Causes

1. **Path Intersection with Obstacles:** Generated trajectories overlapped with occupied map regions due to localization drift
2. **No Valid Path Generation:** Path planner unable to generate feasible routes in corner regions (120,120 and 120,80)

Critical Limitation: Failures clustered in areas with complex obstacle configurations and tight passages where small localization errors cascade into planning failures.

System Limitations Identified

- AMCL algorithm fails in symmetric map regions, confusing robot position with alternative locations
- LiDAR-only approach insufficient for reliable marker detection at long distances or poor visibility angles
- Algorithm fragility in narrow passages and small parking gap scenarios

Planning

Include a detailed breakdown of the planned work and responsibilities of the team members. Update your Gantt chart supported by justification of sequential and parallel tasks. Provide a risk analysis, including technical, cost, and timeline risks, of your main solution and propose at least one alternative approach. Do not forget to include a detailed cost analysis and the deliverables. In addition, discuss all relevant ethical concerns, if applicable. This section should also include all test plans and performance metrics for the sub-systems to be built, integrations plans, and system level test procedures. Please include subsections as required.

Schedule

The Mobile Application Development phase will commence in early February and extend through mid-March, with the communication infrastructure team and software specialists leading the effort. This phase focuses on creating a wireless-based user interface that enables remote interaction with the autonomous vehicle system. The development team will design and implement a mobile-friendly application capable of transmitting high-level commands such as "Park" and "Summon" to the vehicle via Wi-Fi connectivity. The application will incorporate real-time status monitoring features, allowing users to receive feedback regarding vehicle location, operational state, and health system. Command transmission protocols will be established to ensure reliable and responsive communication with minimal latency, meeting the requirement of less than one second delay between user input and vehicle response. Additionally, the application will include emergency stop functionality accessible through both the mobile interface and hardware buttons, ensuring safety during all operational phases.

Running parallel to the Mobile Application Development, the Sensor Integration and Environmental Perception phase will extend from mid-February through late April under the direction of the perception and hardware engineering team. This comprehensive phase encompasses the integration and optimization of multiple sensing modalities, including the ultrasonic scanning system, image processing pipeline, and kalman filter of encoder, IMU and LiDAR scan for perfect localization. The team will develop and refine algorithms for detecting parking spaces, identifying obstacles of minimum 5 centimeters diameter, and processing visual information with 90% accuracy in well-lit environments and 70% accuracy in challenging conditions. As well as LiDAR based navigation to desired parking spot, marker detection algorithms utilizing Aruco markers will be implemented and tested to reliably identify designated parking and recall target locations from various distances and angles.

Simultaneously with the Sensor Integration phase, the Motion Control System Development phase will run from mid-February to early May, led by the control engineering specialists. This phase focuses on developing sophisticated control algorithms that manage vehicle acceleration, deceleration, and steering precision. The team will implement proportional-

integral-derivative feedback control loops utilizing encoder data to achieve accurate wheel rotation and trajectory tracking. Motor driver interfaces will be developed to safely control high-voltage motors using the 3.3-volt outputs from the Raspberry Pi platform. The control system will be tuned to support operation at multiple speed ranges, including standard driving speeds below 0.5 meters per second and precision parking speeds around 0.1 meters per second. System stability analysis and simulation using MATLAB and Simulink will validate controller performance before hardware implementation, ensuring smooth acceleration and braking transitions.

The System Integration and Comprehensive Testing phase will begin in mid-April and continue through early June, bringing together expertise from all technical disciplines. This phase focuses on assembling all developed subsystems—perception, control, communication, and power management—into a unified autonomous vehicle platform. Integrated testing will verify that sensor fusion logic correctly combines camera input with ultrasonic data to make robust navigation decisions. The state machine controlling vehicle behavior through idle, seeking, aligning, parking, and returning states will be validated across multiple operational scenarios. Battery capacity testing will confirm that the onboard power system sustains the Raspberry Pi, microcontroller, DC motor, and servo motors for the required 15-minute continuous operation period. Wireless communication will be evaluated at various distances, with success requiring reliable operation at minimum 3-meter range from the testbed with average command delays below one second.

The Vehicle Prototype Validation and Performance Assessment phase will commence in late May and conclude in mid-June, requiring coordinated evaluation efforts from the entire project team. This phase encompasses comprehensive testing of the complete autonomous parking and vehicle recall system within the 2-meter by 2-meter testbed environment. Parking accuracy will be verified through multiple trials, confirming the vehicle's ability to position itself within the required ± 5 -centimeter error margin in designated 40-centimeter by 20-centimeter parking spaces. Obstacle avoidance capabilities will be tested with static and dynamic obstacles to ensure reliable safety performance. Path planning algorithms will be evaluated for their ability to generate collision-free trajectories and select appropriate parking locations within the specified five-second timeframe. Trajectory tracking accuracy, control system response characteristics including overshoot and steady-state error, and emergency stop latency will be measured and documented. Any discrepancies identified during validation will be resolved through iterative refinement of algorithms and system parameters.

The Final Demonstration and Project Completion phase will occur in late June through early July, representing the culmination of the entire autonomous vehicle development initiative. During this phase, the integrated system will be presented to stakeholders and evaluators, demonstrating the successful achievement of all technical objectives and performance requirements. The presentation will showcase the vehicle's autonomous parking capabilities, obstacle detection and avoidance performance, wireless command responsiveness, and vehicle recall functionality. The team will provide comprehensive documentation including technical specifications, performance test results, system architecture diagrams, and operational procedures. A performance report will be prepared for detailing trajectory tracking accuracy, control response metrics, communication latency measurements, and recommendations for future enhancement. This phase marks the formal conclusion of the development effort and establishes the foundation for potential future refinement or deployment of autonomous parking technology.

Risk Analysis

Technical Risks

- **Localization accuracy failure:**
If LiDAR is used for more than 30 minutes, there will be cumulative error about 5% and it will increase with time. This drift can affect the localization accuracy in long term uses. This can be prevented by adding an automatic reset algorithm for LiDAR at certain time intervals.
- **Sensor fusion failure:**
Encoder quantization errors, IMU bias drift and wheel slip can cause an accumulative pose error which will be mitigated by using filter.
- **Motor control instability:**
There can be oscillations, overshoots, slow responses in speed control that cause a poor navigation accuracy. We will solve this by implementing PID control algorithms for each wheel.

Cost risks

- **Change of the plan:**
If we decide to implement the Plan B, we have to add pi camera to the system. Anticipating this, we created our budget with a margin for it.
- **Power system upgrade:**
If the planned battery pack does not provide sufficient power for the system, more cells will be added. We decided to design a battery pack instead of using a single battery by taking this into account.

Timeline risks

- **Process Delays:**
Integration and calibration of the subsystems are generally expected to be longer than the estimated periods. If the integration of one system takes longer than expected, we will transfer team members working on other systems to this part.
- **Change of the plan:**
Adding camera to the system will cause an extra workload since its fusion with LiDAR takes time. We will focus on the development of only LiDAR solution and improve it as much as possible before camera implementation so that we can add camera as enhancement with clear rollback plan.

Test Plans

Motor Control Test: Given the dynamic path, the car should navigate the desired location. Data communication between sensors and motors will be implemented via RPI Pico controller. The proper operation of the path following by car will be tested by utilizing our navigation test with a fully constructed car system.

Sensor Fusion Test: In the B plan, LiDAR and camera sensors will be fused to help the car to perceive its environment as humans do. Fusion of sensors enables us to develop a 3D model of environment and better sensing of obstacles and more accurate localization, but there is loss from reaction time since the system processes more data in small memory. To be able to test the combined sensors, the already done localization and navigation test will be conducted. Finally, the processing time and accuracy will be compared.

Obstacle Detection and Safety Test: This test will evaluate the ability of the LiDAR sensor to detect obstacles and stop the vehicle safely. Static obstacles with small sizes will be placed

on the path. The vehicle must detect the obstacle in time and stop or avoid it without collision. The stopping distance and system response will be recorded.

Full System Integration Test: This test will combine all subsystems in a single scenario. The vehicle will start, navigate, park, and return to a target location. The overall success rate and system behavior will be evaluated.

Parking Accuracy Test: This test will check whether the vehicle can park inside the target parking area correctly. The vehicle will start from different positions and angles and try to park in a 40 cm × 40 cm space. The final position and orientation error will be measured.

Long-Duration Drift Test: In this test, the vehicle will operate continuously for an extended period (30-60 minutes). Localization error will be measured over time to observe drift effects. The test will help determine how long the system can operate reliably before accuracy decreases.

Summon and Recall Test: In this test, the vehicle will move back to a user-defined location after receiving a summon command. The distance between the final position and the target point will be measured. The test will be repeated from different parking locations to check reliability and consistency.

Power and Endurance Test: This test will evaluate the battery and power system. The vehicle will operate continuously under normal workload. Battery voltage and operating time will be recorded to ensure the system meets the required endurance.

Emergency Stop Test: This test will check the safety of the system. An emergency stop command will be triggered while the vehicle is moving. The vehicle must stop quickly and reliably in all cases. Stopping time and distance will be measured.

Project Deliverables

- **Autonomous Vehicle System:**
A fully working small-scale autonomous vehicle will be delivered. The vehicle can drive in an indoor environment, detect obstacles, park itself in a defined area, and return to a user-defined location using a recall command.
- **Perception and Sensing System:**
The system will include LiDAR, IMU, wheel encoders, and a camera (for the hybrid solution). These sensors will detect obstacles, measure distance, and understand the environment for safe navigation and parking.
- **Localization and Navigation System:**
The vehicle will estimate its position and orientation using LiDAR-based localization. It will plan safe paths and move to parking and recall targets without collision.
- **Mobile User Interface:**
A mobile application will be delivered as the user interface. The app will allow the user to start parking, call the vehicle back, monitor system status, and stop the vehicle in emergencies.

Ethical Concerns

This project includes several ethical concerns that must be considered. Safety is the most important concern, because an autonomous vehicle can cause damage or injury if it moves incorrectly. For this reason, the system must use safe speed limits, reliable LiDAR sensing,

and an emergency stop feature. Another concern is reliability, since LiDAR errors or software bugs can lead to wrong decisions. To reduce this risk, the system is tested carefully in a controlled indoor environment. Privacy is also an ethical issue, even though only LiDAR is used, because system data should not be misused or shared without purpose. Finally, the project should only be used in indoor test areas, because using an experimental autonomous system in public spaces could be unsafe. By addressing these concerns, the project aims to be safe, responsible, and ethical.

Conclusion

This report presented the conceptual design and current development status of the AutoNexa Intelligent Parking and Recall System. Through this project, we designed a small-scale autonomous vehicle capable of navigating in a constrained indoor environment, detecting parking spaces, and performing autonomous parking and recall operations. The proposed system architecture combines perception, localization, navigation, control, and wireless communication subsystems in a modular and scalable manner. The conceptual design demonstrates that autonomous parking can be achieved with high accuracy under realistic engineering constraints.

One of the main outcomes of this project is the validation of LiDAR-based localization and navigation for indoor parking scenarios. Extensive subsystem-level tests showed that the system can achieve acceptable position and orientation accuracy for low-speed autonomous maneuvers. At the same time, the experiments clearly revealed important limitations such as localization drift, reduced performance near boundaries, and sensitivity in narrow regions. These findings provided valuable insight into the practical challenges of autonomous navigation and directly guided our design decisions.

Based on the test results and analysis, we gained a deeper understanding of the importance of sensor fusion and system robustness. While a LiDAR-only solution offers simplicity and lighting-independent operation, it may not be sufficient to meet strict parking accuracy requirements in all scenarios. As a result, an alternative hybrid approach combining LiDAR and vision was proposed to improve reliability and final positioning accuracy. This comparison allowed us to evaluate trade-offs between complexity, performance, and robustness in a structured and objective way.

Overall, this project enabled us to translate theoretical knowledge in control systems, robotics, and embedded software into realistic autonomous system design. We gained practical experience in system decomposition, requirement-driven design, experimental validation, and engineering decision-making based on test data. The conceptual design presented in this report establishes a strong foundation for full system integration and future improvements, and it provides a clear roadmap toward a reliable and accurate autonomous parking prototype.

Appendix 1: Weighted Objective Tree

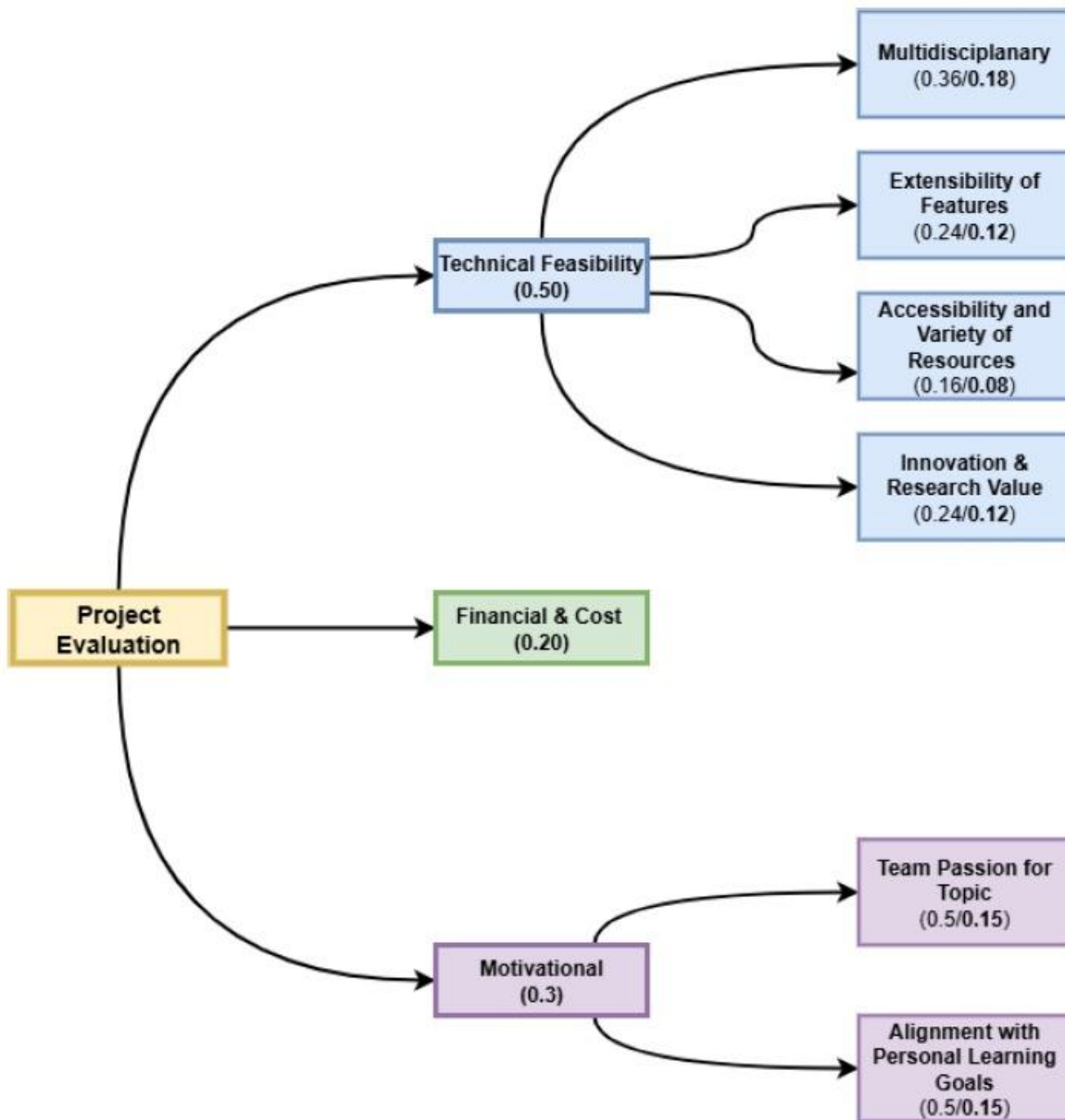


Figure 14. Weighted objective tree used for system design decisions.

Appendix 2: Gantt Chart

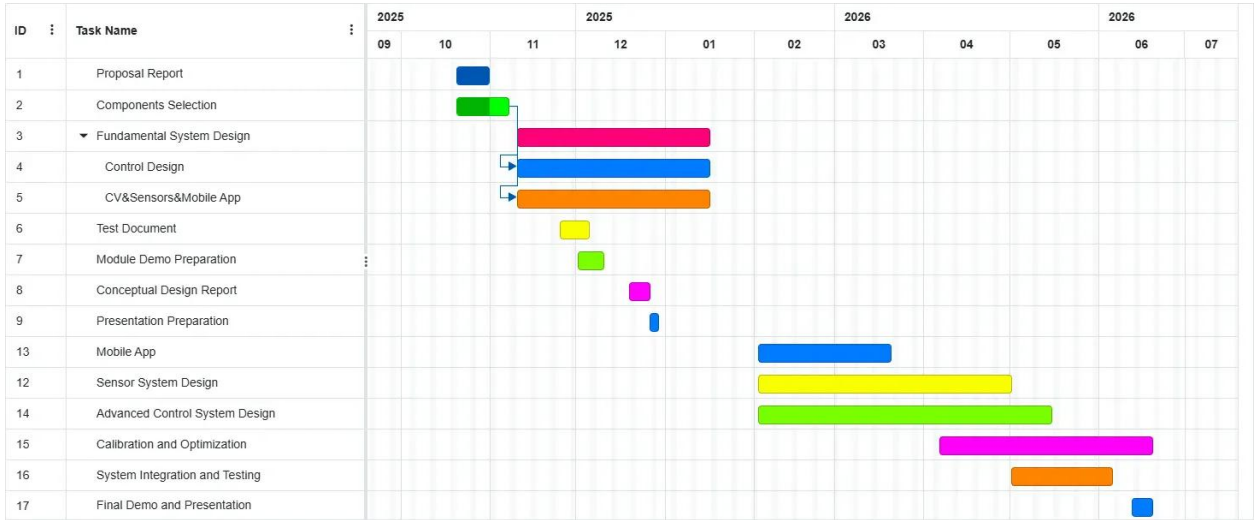


Figure 15. Gantt Chart of The Tasks to Be Performed

Appendix 3: Test Results

Localization Test Results

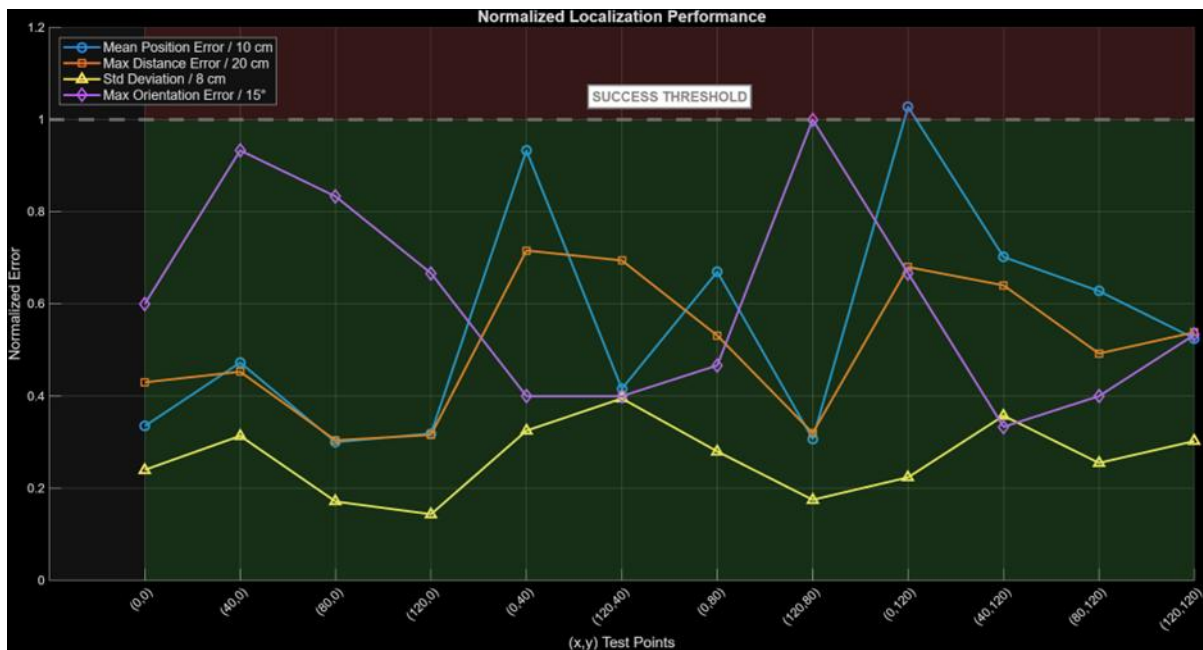


Figure 16. Normalized Localization Performance with Respect to Acceptance Criteria

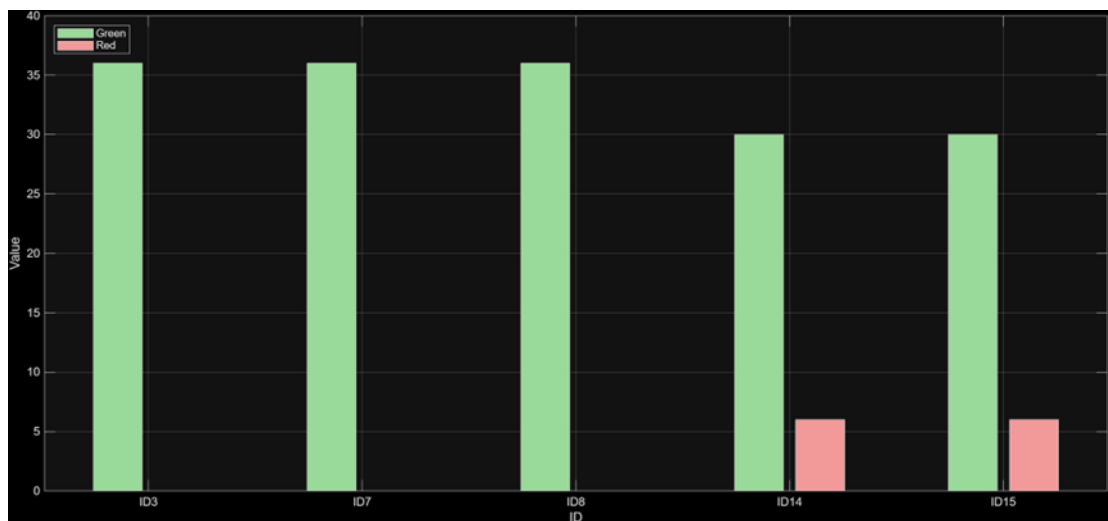


Figure 17. Spatial Distribution of Navigation Success and Failure Cases.

Table 4. Localization Accuracy Metrics per Test Point.

Test Point (x,y)	Mean Error (cm)	Variance (cm ²)	Std. Dev. (cm)	CI 95% (cm)	Max Distance Error (cm)
(0,0)	3.35	3.70	1.92	0.81	8.60
(40,0)	4.73	6.29	2.51	1.06	9.06
(80,0)	3.00	1.87	1.37	0.58	6.08
(120,0)	3.19	1.32	1.15	0.49	6.32
(0,40)	9.33	6.78	2.60	1.10	14.32
(120,40)	4.16	9.98	3.16	1.33	13.89
(0,80)	6.70	5.01	2.24	0.95	10.63
(120,80)	3.07	1.96	1.40	0.59	6.40
(0,120)	10.28	3.19	1.79	0.75	13.60
(40,120)	7.02	8.20	2.86	1.21	12.81
(80,120)	6.28	4.18	2.04	0.86	9.85
(120,120)	5.25	5.87	2.42	1.02	10.77

Table 5. Orientation (Heading) Error Statistics per Test Point.

Test Point (x,y)	Mean Angle Error (deg)	Angle Variance (deg ²)	Angle Std. Dev. (deg)	CI 95% (deg)	Max Angle Error (deg)
(0,0)	2.93	4.39	2.10	1.75	9.00
(40,0)	2.46	3.20	1.79	1.50	14.00
(80,0)	2.95	8.96	2.99	2.50	12.00
(120,0)	1.99	2.64	1.62	1.36	10.00
(0,40)	2.58	2.47	1.57	1.31	6.00
(120,40)	3.17	2.83	1.68	1.41	6.00
(0,80)	3.08	0.94	0.97	0.81	7.00
(120,80)	4.56	8.96	2.99	2.50	15.00
(0,120)	3.33	3.21	1.79	1.50	10.00
(40,120)	3.08	0.79	0.89	0.74	5.00
(80,120)	2.17	1.84	1.36	1.13	6.00
(120,120)	3.08	1.87	1.37	1.14	8.00

Navigation Test Results

Table 6. Distance Measurement Results (ID = 8).

x (cm)	y (cm)	Mean (cm)	Variance (cm²)	Std. Dev. (cm)	CI 95% (cm)
0	0	4.08	0.0183	0.1353	0.3360
0	40	4.81	0.1159	0.3404	0.8457
0	80	7.72	0.0632	0.2515	0.6247
0	120	4.42	0.0513	0.2265	0.5626
40	120	6.63	0.0680	0.2608	0.6479
80	120	7.41	0.0819	0.2862	0.7109
120	120	7.93	0.1069	0.3270	0.8122
120	80	5.37	0.1057	0.3252	0.8078
120	40	6.55	0.1414	0.3761	0.9342
120	0	4.19	0.1125	0.3355	0.8333
80	0	5.46	0.0996	0.3156	0.7841
40	0	6.02	0.1024	0.3200	0.7949

Table 7. Distance Measurement Results (ID = 7).

x (cm)	y (cm)	Mean (cm)	Variance (cm²)	Std. Dev. (cm)	CI 95% (cm)
0	0	4.84	3.37	1.84	4.56
0	40	5.71	3.59	1.89	4.71
0	80	5.18	1.13	1.06	2.64
0	120	5.19	4.54	2.13	5.29
40	120	5.75	1.62	1.27	3.16
80	120	6.22	3.88	1.97	4.89
120	120	5.19	1.49	1.22	3.03
120	80	5.60	4.36	2.09	5.19
120	40	6.23	2.39	1.55	3.84
120	0	5.16	1.94	1.39	3.46
80	0	5.79	3.84	1.96	4.87
40	0	6.19	1.86	1.36	3.39

Table 8. Distance Measurement Results (ID = 13).

x (cm)	y (cm)	Mean (cm)	Variance (cm²)	Std. Dev. (cm)	CI 95% (cm)
0	0	4.83	1.42	1.19	2.96
0	40	5.70	2.57	1.60	3.98
0	80	5.78	3.44	1.85	4.61
0	120	5.73	3.81	1.95	4.85
40	120	5.25	1.36	1.17	2.89
80	120	6.00	2.87	1.69	4.21
120	120	5.63	3.63	1.91	4.74
120	80	5.63	1.68	1.30	3.22
120	40	5.00	2.90	1.70	4.23
120	0	5.76	3.13	1.77	4.40
80	0	5.36	1.13	1.07	2.65
40	0	6.16	1.98	1.41	3.50

Table 9. Distance Measurement Results (ID = 14).

x (cm)	y (cm)	Mean (cm)	Variance (cm²)	Std. Dev. (cm)	CI 95% (cm)
0	0	4.75	2.22	1.49	3.70
0	40	5.69	2.92	1.71	4.24
0	80	5.44	1.32	1.15	2.85
0	120	5.07	3.20	1.79	4.45
40	120	5.73	1.29	1.14	2.82
80	120	6.15	3.34	1.83	4.54
120	120	–	–	–	–
120	80	–	–	–	–
120	40	6.07	2.00	1.42	3.52
120	0	5.26	1.74	1.32	3.28
80	0	5.70	3.26	1.81	4.49
40	0	6.21	1.98	1.41	3.49

Table 10. Distance Measurement Results (ID = 15).

x (cm)	y (cm)	Mean (cm)	Variance (cm²)	Std. Dev. (cm)	CI (95%)
0	0	4.88	2.55	1.60	3.96
0	40	5.82	2.53	1.59	3.95
0	80	5.80	2.72	1.65	4.09
0	120	5.69	3.15	1.78	4.41
40	120	5.23	1.20	1.09	2.72
80	120	6.16	2.61	1.62	4.01
120	120	–	–	–	–
120	80	–	–	–	–
120	40	5.04	2.54	1.59	3.96
120	0	5.63	3.00	1.73	4.30
80	0	5.41	1.25	1.12	2.78
40	0	6.10	1.58	1.26	3.12